

Plaque Radiotherapy for Choroidal Melanoma Encircling the Optic Disc (Circumpapillary Choroidal Melanoma)

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Objective: To report results of plaque radiotherapy for choroidal melanoma that completely encircles the optic disc (circumpapillary melanoma).

Methods: Retrospective medical record review over a 31-year period of 37 consecutive patients. The main outcome measures were treatment complications, long-term visual acuity, enucleation, tumor recurrence, metastasis, and death.

Results: The median patient age at treatment was 69 years (range, 20-86 years). The presenting complaint was visual loss in 19 eyes (51%), photopsia in 5 (14%), and visual field loss in 3 (8%). All tumors touched and encircled the optic disc for 360°. The quadrant location of the main tumor epicenter was superior in 8 eyes (22%), nasal in 10 (27%), inferior in 9 (24%), and temporal in 10 (27%). The median tumor basal diameter was 11 mm (range, 4.8-20 mm) and median tumor thickness was 3.6 mm (range, 1.8-14.8 mm). The optic disc was obscured to some extent by overhanging tumor in 19 cases (52%). The most commonly used isotope for plaque radiotherapy was iodine 125 (n=34 cases; 92%), and a notched plaque design was used in 34 cases (92%). Planned adjunctive treatment included transpupillary thermotherapy in 17 cases (49%) and argon laser photocoagulation in 6 of 35 cases (17%) with follow-up. Of the 28 eyes with more than 5 months' follow-up (mean, 52 months; median, 46 months; range, 5-234 months), treatment complications included nonproliferative and proliferative retinopathy in 11 (39%) and 7 eyes (25%); maculopathy in 7 (25%); papillopathy in 9 eyes (32%); neovascular glaucoma in 5 (18%); and vitreous hemorrhage in 13 (46%). Pars plana vitrectomy was required in only 2 of 13 eyes (15%) with persistent vitreous hemorrhage. Long-term visual acuity of 20/200 or worse was observed in 13 eyes (62%), and 12 eyes (57%) lost more than 5 Snellen visual acuity lines, excluding 7 cases (25%) in which enucleation was necessary. Recurrence was noted in 4 cases (14%), of which 3 were treated with enucleation and 1 with transpupillary thermotherapy. Systemic metastasis occurred in 1 patient (4%) and there were no melanoma-specific deaths during the study period.

Conclusions: Custom-designed plaque radiotherapy is a potential treatment for selected patients with circumpapillary choroidal melanoma. We found satisfactory local tumor control, and globe retention was achieved in 75% of eyes with more than 5 months' follow-up.

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THE MANAGEMENT OF CHOROIDAL melanoma located adjacent to or touching the optic disc (juxtapapillary choroidal melanoma) is challenging.¹⁻³ Treatment modalities include enucleation,⁴ transpupillary thermotherapy (TTT),^{5,6} proton beam irradiation,⁷ stereotactic radiosurgery,⁸ and plaque radiotherapy.⁹ Previous reports from our group have shown plaque radiotherapy to be an equivalent option to enucleation for selected juxtapapillary melanoma.^{10,11}

Little attention has been paid in the literature to choroidal melanoma that completely encircles the optic nerve (circumpapillary choroidal melanoma). In a

clinicopathological case report of an enucleated eye, deVeer¹² described a melanoma surrounding the optic disc and refuted the idea that such lesions arose within the nerve itself. It is likely that other similar eyes have come to enucleation in the past.¹³ In most circumstances, enucleation is the treatment of choice for circumpapillary melanoma; however, in certain situations where enucleation is not desirable, such as small tumor size, good vision in the involved eye, and poor vision in the opposite eye, custom-designed plaque radiotherapy has been used. In this retrospective case series, we present the 31-year experience of the Ocular Oncology Service at Wills Eye Hospital, Philadelphia, Pennsylvania, detailing the

tumor features and radiation dosimetry parameters, treatment complications, and tumor control for circumpapillary choroidal melanomas.

METHODS

The electronic database of patients seen at the Ocular Oncology Service of Wills Eye Hospital was searched for patients with circumpapillary choroidal melanoma treated with plaque radiotherapy. Circumpapillary choroidal melanoma was defined as a choroidal melanoma that was within 1 mm of the optic disc and surrounded the optic disc for all 12 clock hours (**Figures 1, 2, and 3**). Those tumors that completely overhung the optic nerve were included if there was ophthalmoscopic, angiographic, or ultrasonographic evidence of the entire choroidal circumference of the optic nerve being involved. (Figure 2B-D). The study comprised 37 consecutive patients evaluated between October 1974 and May 2005.

Our methods of clinical examination, diagnosis, and treatment have been described previously.¹ For the purposes of this case series, we recorded the following data prior to initial therapy: age, race, sex, medical history (hypertension, diabetes mellitus, hypercholesterolemia, dysplastic nevus syndrome, skin melanoma), family history (choroidal or skin melanoma), ocular symptoms, examination findings (visual acuity, anterior segment details, intraocular pressure), and tumor findings (quadrant of tumor epicenter, number of clock hours surrounding the optic disc, percentage of overhang of the optic disc by the tumor, distance of tumor margin to foveola and optic disc in millimeters, largest basal diameter and thickness in millimeters, tumor configuration, color, extraocular extension, retinal invasion, subretinal fluid, orange pigment, optic disc swelling, and retinal vascular congestion). The following plaque radiation particulars were recorded: radionuclide isotope (iodine 125 [¹²⁵I], ruthenium 106, cobalt 60 [⁶⁰Co], iridium 192), plaque size in millimeters, plaque shape (round, notched, postage stamp), custom seed distribution (full or posterior, according to the basal dimensions and elevation of the tumor and oriented perpendicular to the optic nerve and foveola to minimize radiation exposure to these sensitive structures), time of radiation exposure (hours), and radiation dose (centigrays) and rate (centigrays per hour) to the tumor apex, tumor base, optic disc, foveola, and lens. Currently, 8000 cGy is the preferred radiation dose to the tumor apex. Adjunctive treatment details with TTT,¹⁴ argon laser photocoagulation (ALPC), or krypton laser photocoagulation were ascertained. Cumulative follow-up data included treatment complications (any degree of nonproliferative and proliferative radiation retinopathy, maculopathy, papillopathy, radiation cataract, neovascular glaucoma, vitreous hemorrhage, scleral necrosis, venous or arterial occlusion), long-term visual acuity, loss of vision of more than 5 Snellen visual acuity lines, enucleation, tumor recurrence (Figure 3), systemic metastases, and death.

RESULTS

Circumpapillary choroidal melanoma was diagnosed clinically in 37 patients. **Table 1** details the patient demographics. The tumor features are presented in **Table 2** (n=37). The tumor epicenter was equally distributed among different quadrants. The anterior tumor margin was behind the equator in 30 eyes (81%), and in 2 eyes (5%), the tumor extended to the ciliary body. There were 3 eyes (8%) with subtle optic disc swelling. Retinal invasion and retinal vascular congestion were found in 1 eye each (3%).

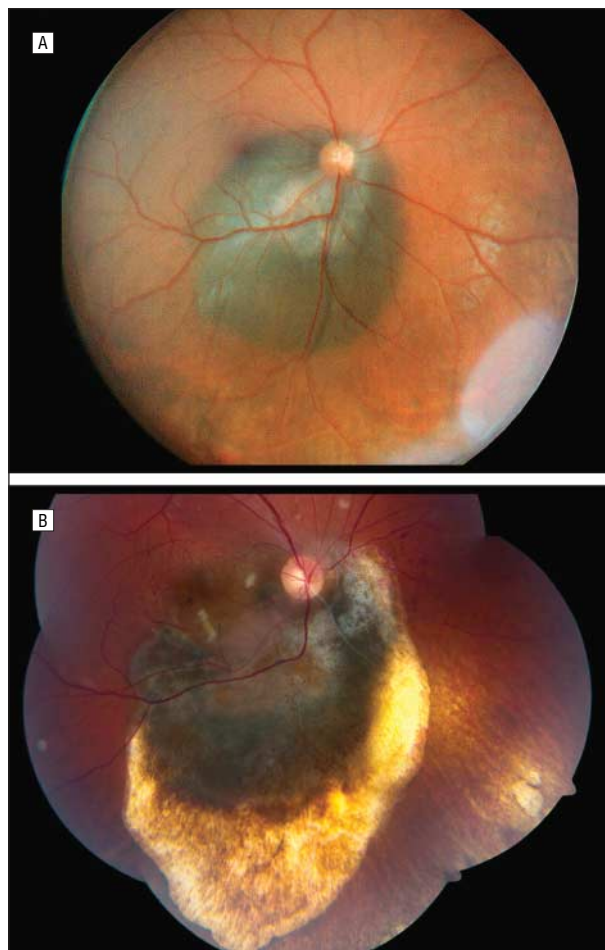


Figure 1. Circumpapillary melanoma in a 20-year-old man before and after plaque radiotherapy. A, Before plaque radiotherapy. The visual acuity was 20/30 and the melanoma was treated with iodine 125 notched plaque with posterior distribution and 3 sessions of transpupillary thermotherapy (TTT). B, After plaque radiotherapy. Thirteen months following plaque radiotherapy combined with TTT, the tumor appeared regressed. Note the nerve fiber layer infarctions and sclerosed blood vessels. The visual acuity at 13 months' follow-up was 20/70.

Table 3 lists the radiotherapy characteristics for the 37 eyes. The radiation dose to the tumor apex was a median of 8000 cGy and to the tumor base was 30 600 cGy. An apex dose lower than 8000 cGy was used in 4 early cases (three ⁶⁰Co plaques at doses of 4500 cGy, 6000 cGy, and 6020 cGy, and one ¹²⁵I plaque at 7600 cGy). Iodine 125 was the most frequently used isotope (34 cases; 92%) and 34 plaques (92%) were deep notched, with radioactive seeds arranged on either side of the notch in such a way as to allow delivery of an adequate tumoricidal dose to the entire tumor. The seeds were placed in a posterior distribution to minimize radiation exposure to normal tissue in 9 patients (24%) with relatively smaller tumor basal diameter. There were 3 early cases treated with round plaques with a full distribution of radioactive seeds that delivered an adequate dose to the entire tumor but also delivered an average of 37159 cGy to the optic disc.

There were 35 cases completing more than 3 months' follow-up. Following plaque radiotherapy, 11 patients (31%) did not receive any adjunctive therapy (non-TTT group), 6 (17%) had ALPC (ALPC group), and 17 (49%)

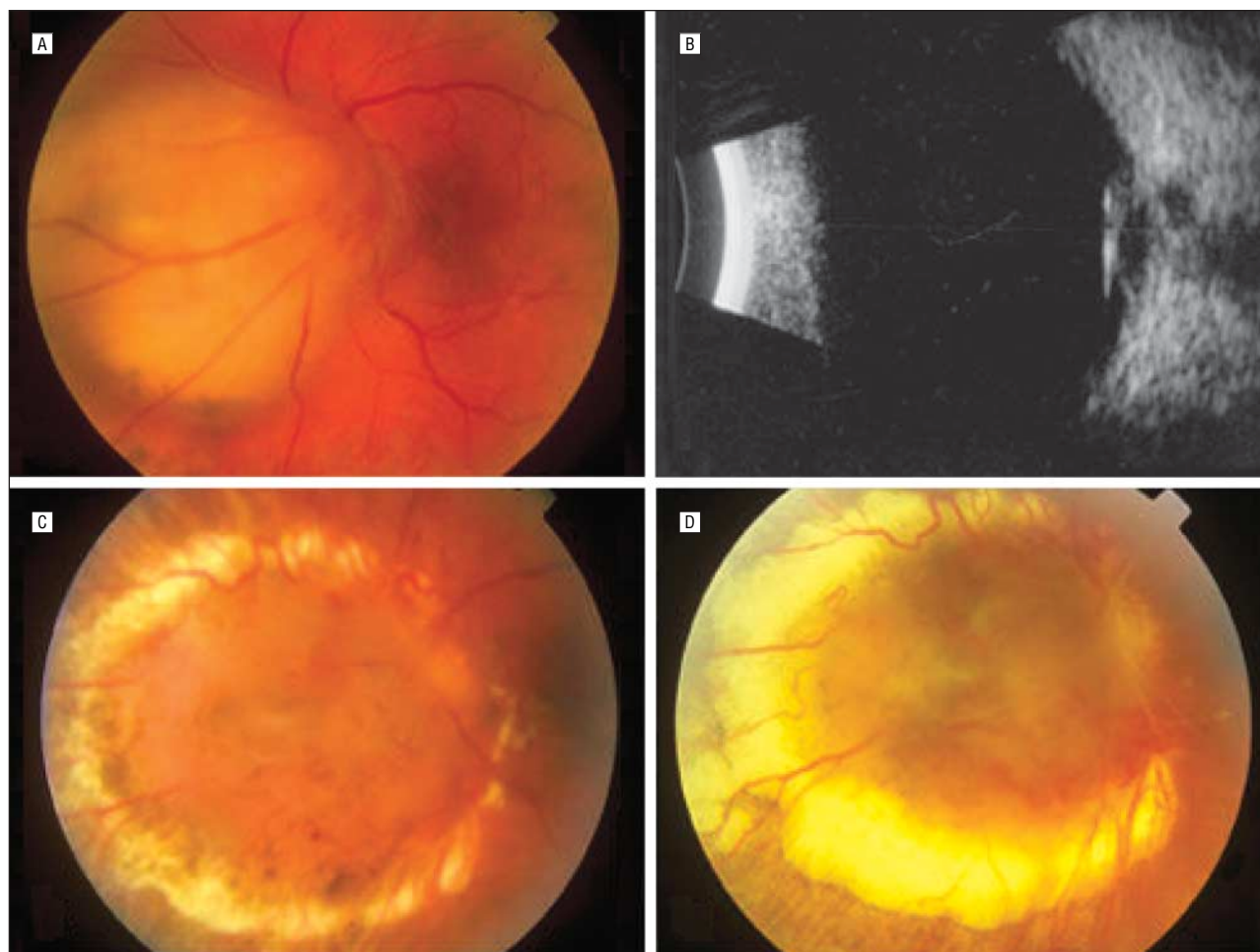


Figure 2. Circumpapillary melanoma overhanging the optic nerve in a 61-year-old man before and after plaque radiotherapy. A, Before plaque radiotherapy. The amelanotic circumpapillary melanoma completely overhung the optic disc and visual acuity was 20/30. The melanoma was treated with full-distribution notched iodine 125 plaque and 3 sessions of argon laser photocoagulation. B, Before plaque radiotherapy. B-scan ultrasonography showed an acoustically hollow tumor overhanging the optic nerve. C and D, After plaque radiotherapy. At 18 months (C) and 76 months (D), the tumor was regressed. The final visual acuity was no light perception.

had up to 3 sessions of TTT (TTT group) (**Table 4**). There was 1 patient excluded from the outcomes analysis who initially received TTT alone and then had plaque radiotherapy for recurrence. The tumors in the non-TTT group (n=11) had a median basal diameter of 13 mm (range, 5-20 mm) and a median thickness of 6.0 mm (range, 2.5-14.8 mm) and 1 case (9%) was a diffuse melanoma. The TTT group (n=17) had a median basal diameter of 10 mm (range, 6-16 mm) and a median thickness of 3.3 mm (range, 2.2-5.0 mm) and 7 cases (41%) were diffuse melanoma. The median radiation dose to the apex in the TTT group was 8000 cGy (range, 8000-10 500 cGy) and in the non-TTT group, 8110 cGy (range, 4500-10 500 cGy).

The major outcomes in 28 eyes with a minimum of 5 months' follow-up (mean, 52 months; median, 46 months; range, 5-234 months) are listed in **Table 5** and were subdivided into the non-TTT (n=11) and TTT (n=17) groups. Nonproliferative radiation retinopathy was found in 11 cases (39%), of which 4 cases (36%) were in the non-TTT group and 7 cases (41%) were in the TTT group. Proliferative radiation retinopathy occurred in 7 cases (25%) overall, though a higher proportion occurred in

the non-TTT group (5 cases; 45%) compared with the TTT group (2 cases; 12%). There were 7 cases (25%) in the whole series with maculopathy; all of these were in the TTT group. Papillopathy, defined as optic disc edema or pallor not attributable to any other cause except radiation, was found in 9 cases (32%), with a lower proportion in the non-TTT group (2 cases; 18%) than in the TTT group (7 cases; 41%). Radiation cataract developed at a median time of 23 months in 10 eyes (45%), with 5 eyes (63%) in the non-TTT group and 5 eyes (36%) in the TTT group. Neovascular glaucoma occurred in 5 cases (18%), of which 4 (36%) were in the non-TTT group and only 1 (6%) in the TTT group. Vitreous hemorrhage developed at a median of 17 months in 13 cases (46%), though 9 (81%) of these were in the non-TTT group and 4 (24%), in the TTT group. Pars plana vitrectomy was required in 2 of 13 eyes (15%) with persistent dense vitreous hemorrhage, and conservative management alone was sufficient in the remaining 11 eyes. There were no cases of scleral necrosis in either group. Branch retinal vein occlusion was noted in 9 cases (32%) and branch retinal artery occlusion, in 6 cases (21%). All branch vascular occlusions occurred in the TTT group. There were

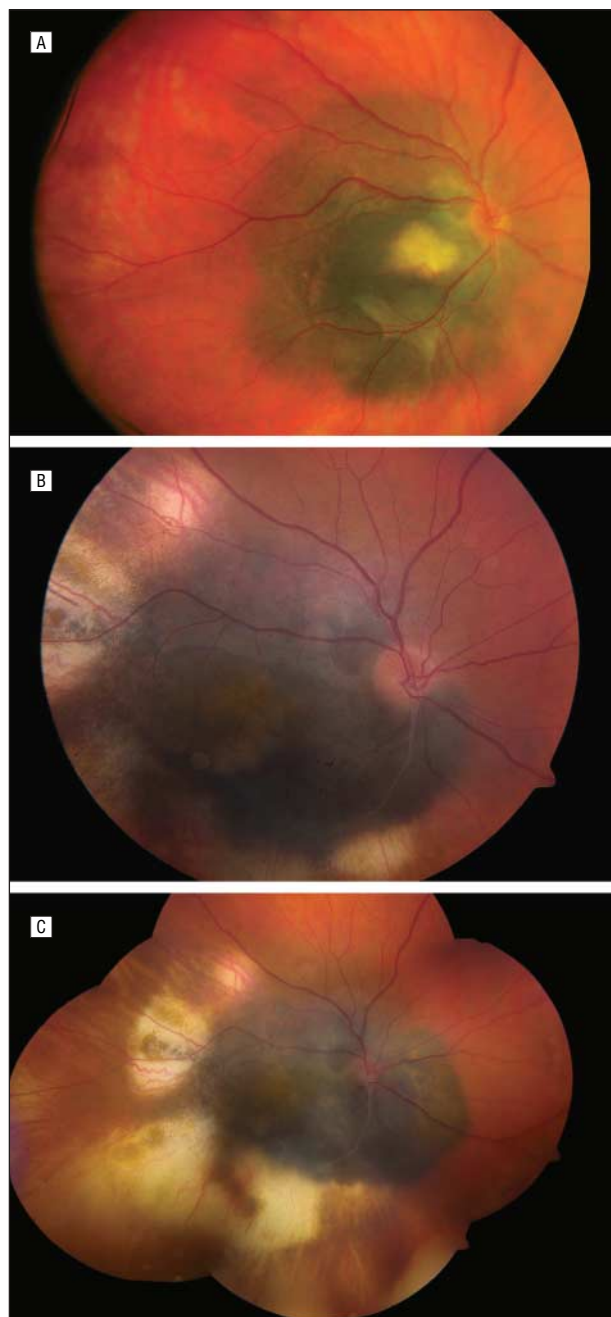


Figure 3. Circumpapillary melanoma with recurrence following plaque radiotherapy in a 65-year-old man (case 2, Table 6). A, Before plaque radiotherapy. The melanoma was treated with iodine 125 notched plaque with full distribution and 3 sessions of transpupillary thermotherapy. Pretreatment visual acuity was 20/200. B, After plaque radiotherapy. Tumor regression is noted at 27 months. C, After plaque radiotherapy. Tumor recurrence in the macular region was found at 34 months, requiring enucleation.

no cases of central retinal vascular occlusion and there were no cases of direct thermal damage to the optic nerve with TTT, as judged by immediate loss of vision or by subsequent rapid onset of optic disc edema.

Long-term visual acuity was assessed as final visual acuity and visual loss more than 5 Snellen lines in the 28 eyes with more than 5 months' follow-up. Excluding enucleated eyes, long-term visual acuity of 20/200 or worse was found in 13 eyes (62%). In the non-

Table 1. Demographics of 37 Patients With Circumpapillary Choroidal Melanoma Treated With Plaque Radiotherapy

Features	No. (%) (n = 37)
Age, y, median (mean [range])	69 (63 [20-86])
Race	
White	37 (100)
African American	0
Hispanic	0
Asian	0
Mixed	0
Sex	
M	24 (65)
F	13 (35)
Medical history ^a	
Hypertension	14 (38)
Diabetes mellitus	5 (14)
Hypercholesterolemia	3 (8)
Dysplastic nevus syndrome	0
Skin melanoma	0
Family history	
Choroidal melanoma	0
Skin melanoma	2 (5)
Neither	35 (95)
Ocular symptoms	
Decreased visual acuity	19 (51)
Flashes/floaters	5 (14)
Visual field defect	3 (8)
Pain	0
Diplopia	0
Asymptomatic	10 (27)

^aSome patients had more than 1 condition.

TTT group, 8 eyes (100%) had a visual acuity of 20/200 or worse, whereas 5 eyes (38%) reached this level in the TTT group. Visual loss of more than 5 Snellen lines occurred in 12 eyes (57%), of which 6 (75%) were in the non-TTT group and 6 (46%) in the TTT group.

Enucleation was necessary in 7 (25%; n=28) of the patients in this study. Enucleation rates were similar in both the non-TTT group (3 eyes; 27%) and the TTT group (4 eyes; 24%). Tumor recurrence (3 cases), neovascular glaucoma (3 cases), and severe ocular pain (1 case) were the reasons for enucleation.

Tumor recurrence was found in 4 cases (14%), all of which were in the TTT group (24%). The 4 recurrent cases are summarized in **Table 6**. The tumor features of these cases were remarkable for the following: 2 of 4 of these lesions were diffuse melanoma, all had orange pigment and subretinal fluid, and 2 had optic disc swelling. They had all received notched plaque radiotherapy and adjuvant TTT as the primary treatment. The interval between plaque radiotherapy and recurrence was a range of 10 to 34 months. Tumor control in 1 case was achieved with further TTT and the remaining 3 cases underwent enucleation. In this series, 19 (52%) of the tumors overhung the optic disc to some extent and only 1 of these manifested recurrence (Table 6, case 4).

Metastasis developed at 66 months in 1 patient (4%) who was in the non-TTT group (9%). There were no deaths recorded in the study period.

Table 2. Tumor Features of 37 Patients With Circumpapillary Choroidal Melanoma Treated With Plaque Radiotherapy

Features	No. (%) (n = 37)
Tumor meridian	
Superior	8 (22)
Nasal	10 (27)
Inferior	9 (24)
Temporal	10 (27)
Proximity to foveola, mm, median (mean [range])	0.3 (1 [0-4.5])
Tumor base, mm, median (mean [range])	11 (11 [4.8-20])
Tumor thickness, mm, median (mean [range])	3.6 (4.8 [1.8-14.8])
Thickness >2 mm	35 (95)
Subretinal fluid	27 (73)
Orange pigment	21 (57)
Tumor shape	
Plateau	18 (49)
Dome	9 (24)
Mushroom	10 (27)
Color	
Pigmented	20 (54)
Partially pigmented	9 (24)
Nonpigmented	8 (22)
Growth pattern	
Circumscribed	26 (70)
Diffuse	11 (30)
Anterior limit of tumor	
Macula	0
Equator	30 (81)
Ora serrata	5 (14)
Ciliary body	2 (5)
Proximity to optic disc, mm, median (mean [range])	0 (0 [0-0])
Overhang of optic nerve, %	
0	18 (49)
1-50	4 (11)
51-100	15 (41)
Optic disc swelling	3 (8)
Associated features	
Extraocular extension	1 (3)
≤ 3 mm	1 (3)
> 3 mm	0
Retinal invasion	1 (3)
Retinal vascular congestion	1 (3)

COMMENT

In this report, we evaluate the clinical features and treatment outcomes of eyes with choroidal melanoma completely encircling the optic disc (circumpapillary choroidal melanoma) and treated with plaque radiotherapy. Juxtapapillary melanoma is a variant of choroidal melanoma located within 1 mm of the optic disc margin and has a previously estimated prevalence of less than 10% of all posterior uveal melanomas.¹⁰ In our current experience, juxtapapillary choroidal melanoma represents approximately 16% of patients treated with plaque radiotherapy (C.L.S. and J.A.S., unpublished data, August 2006). Circumpapillary melanoma is even more unusual, with only 37 cases (<1% of more than 4000 plaque treatments) managed with plaque radiotherapy at the Ocu-

Table 3. Plaque Radiotherapy Parameters of 37 Patients With Circumpapillary Choroidal Melanoma Treated With Plaque Radiotherapy

Features	No. (%) (n = 37)
Radionuclide	
Iodine 125	34 (92)
Ruthenium 106	0
Cobalt 60	3 (8)
Iridium 192	0
Plaque shape	
Round ^a	3 (8)
Notched	34 (92)
Postage stamp	0
Distribution area, mm	
Full	28 (76)
Posterior	9 (24)
Plaque size, mm, median (mean [range])	15 (16 [15-22])
Radiation duration, h, median (range)	98 (80-236)
Dose, cGy, median (range)	
Apex	8000 (4500-12 500)
Base	30 600 (10 900-49 800)
Optic disc	10 878 (1358-62 055)
Foveola	12 853 (3036-34 350)
Lens	1174 (564-8076)
Rate, cGy/h, median (range)	
Apex	84 (23-128)
Base	229 (114-506)
Optic disc	123 (45-418)
Foveola	128 (30-299)
Lens	11 (5-57)

^aOne round plaque was cobalt 60.

Table 4. Adjunctive Treatment Characteristics of 35 Cases With Circumpapillary Choroidal Melanoma Treated With Plaque Radiotherapy With Follow-up^a

Features	No. (%) (n = 35)
After brachytherapy adjunctive treatment	
ALPC	6 (17)
TTT	17 (49)
None	11 (31)
Total No. of TTT treatments, session	
1	3 (18)
2	3 (18)
3	11 (65)
Location of TTT treatment	
Extrafoveal	13 (76)
Foveal	4 (24)

Abbreviations: ALPC, argon laser photocoagulation; TTT, transpupillary thermotherapy.

^aThere was 1 patient who was treated by primary TTT, then had plaque radiotherapy for recurrence. This case was excluded from this table and the outcomes summary in Table 5.

lar Oncology Service at Wills Eye Hospital over 31 years. It is likely that tumor initiation occurs in 1 juxtapapillary sector, followed by circumferential enlargement until the entire optic disc is surrounded. In the current study, we addressed this more advanced subset of melanoma that completely surrounded the optic disc.

There is little information in the literature on management of circumpapillary choroidal melanoma. A Medline search with the key words "circumpapillary" and "choroidal melanoma" from 1966 to June 2006 revealed no references on therapeutic outcomes. Because of the technical difficulty of eye-sparing treatment techniques at this site,¹⁵ most eyes with circumpapillary choroidal melanoma are managed with enucleation. In fact, juxta-papillary melanoma that circumscribes more than 180° of the optic disc is usually managed with enucleation.¹⁶ Plaque radiotherapy for juxta-papillary melanoma has been previously evaluated⁹⁻¹¹ but this is the first series, to our knowledge, in which circumpapillary choroidal melanoma has been evaluated. The Collaborative Ocular Melanoma Study (COMS) excluded lesions that were closer than 2 mm from the optic disc center and subtended an angle greater than 90° for entry into their prospective randomized controlled trial of plaque radiotherapy for medium-sized choroidal melanoma.^{17,18} Previous retrospective series of juxta-papillary melanoma managed with enucleation vs plaque radiotherapy showed equivalent survival rates in the 2 groups and satisfactory tumor control in the plaque radiotherapy group.¹⁰

In our series, the decision to treat these cases with plaque radiotherapy rather than enucleation was based on thinner tumor size, good potential vision in the affected eye, poor vision in the opposite eye, and ultimate patient preference after detailed counseling. The majority of tumors were treated with a custom-designed notched ¹²⁵I plaque.¹⁹ A deep notch to house the optic nerve and its surrounding sheaths, with precisely aligned radioactive seed placement in the arms of the notch, allows irradiation of the entire circumpapillary tumor. A similar principle can be applied to a round plaque, but our preference for the notched plaque is based on the precision and security of placement at the nerve, as well as improved dosimetry in the unshielded notched area of the plaque. The challenge in plaque design and placement cannot be overlooked because this requires deep dissection into the peri-optic tissues of the orbit, precise tumor localization with scleral depression, and posterior placement and suturing of the plaque in the postequatorial sclera. Though intraoperative ultrasonography can be used as an adjunctive step in aligning the plaque,^{20,21} this technique was not routinely used in the cases reported herein. All plaques were at least 15 mm in diameter, and in 24%, the seed arrangement was posteriorly weighted to spare normal equatorial retina. It might be tempting to use 2 sequential notched plaques, oriented at 180° to each other to cover the area of the circumpapillary tumor. However, this would increase the cumulative radiation exposure to the optic nerve and foveola, resulting in greater radiation complications. Whether that treatment option is equivalent in tumor control to our custom plaque design, with radioactive seed orientation to cover the notched area, remains to be determined and is not the subject of the present study.

One of our current methods for treatment of posterior uveal melanoma includes the combination of plaque radiotherapy followed by adjuvant TTT.¹⁴ To evaluate the role of TTT in the management of circumpapillary choroidal melanoma, we divided our patients into TTT and

Table 5. Summary of Outcomes by TTT Group in 28 Cases With Circumpapillary Choroidal Melanoma Treated With Plaque Radiotherapy With Follow-up

Outcomes	No. (%)		
	Plaque, No TTT (n = 11)	Plaque and TTT (n = 17)	Overall (n = 28)
Complications			
Retinopathy, nonproliferative	4 (36)	7 (41)	11 (39)
Retinopathy, proliferative	5 (45)	2 (12)	7 (25)
Maculopathy	0	7 (41)	7 (25)
Papillopathy	2 (18)	7 (41)	9 (32)
Cataract ^a	5 (63)	5 (36)	10 (45)
Neovascular glaucoma	4 (36)	1 (6)	5 (18)
Vitreous hemorrhage	9 (81)	4 (24)	13 (46)
Scleral necrosis	0	0	0
Major branch retinal vein occlusion	0	9 (53)	9 (32)
Major branch retinal artery occlusion	0	6 (35)	6 (21)
Central retinal vein or artery occlusion	0	0	0
Visual acuity			
Final visual acuity ≤ 20/200 ^b	8 (100)	5 (38)	13 (62)
Visual loss (> 5 Snellen lines) ^b	6 (75)	6 (46)	12 (57)
Enucleation, all causes	3 (27)	4 (24)	7 (25)
Tumor recurrence	0	4 (24)	4 (14)
Metastasis, all cases	1 (9)	0	1 (4)
Death, melanoma specific	0	0	0
Mean follow-up, mo, mean (median [range])	77 (57 [31-234])	36 (20 [5-90])	52 (46 [5-234])

Abbreviation: TTT, transpupillary thermotherapy.

^aExcludes previous pseudophakia and nonradiation cataract cases (3, 3, and 6 cases, respectively).

^bExcludes enucleations (3, 4, and 7 cases, respectively).

non-TTT subgroups and compared outcomes and complications in these 2 groups at a mean follow-up of 52 months (Table 5). Although we recognize the limitations of this approach in this small series, it still yields a useful comparison. Nonproliferative retinopathy affected the same proportion of eyes in the non-TTT and TTT groups, but proliferative retinopathy was more frequent in the non-TTT group (45% vs 12%). This difference could be attributed to a different radiation dose to the disc because tumors in the non-TTT group had greater mean thickness; however, the radiation dose to the disc in the non-TTT group was in fact slightly lower than in the TTT group (10 055 cGy and 11 703 cGy, respectively) and the radiation dose to the foveola was similar in the 2 groups (11 247 cGy and 11 530 cGy). It is likely that the difference in proliferative retinopathy is related to a longer follow-up period in the non-TTT group (77 months) compared with the TTT group (36 months). One could also speculate that TTT might induce complete retinal laser damage on ischemic retina and minimize proliferative retinopathy. No cases of maculopathy and fewer eyes with papillopathy were found in the non-TTT group, whereas each of these complications occurred in 41% of the TTT group. Potential reasons for this include underdiagnosis of maculopathy in the non-TTT group because of hazy media from cataract or vitreous hemorrhage and overdiagnosis in the TTT group because of higher rates of branch retinal vascular occlusion. Radiation cataract, neovascular glaucoma, and vitreous hemorrhage developed in a greater proportion of the

Table 6. Features of 4 Cases With Recurrence of 37 Patients With Circumpapillary Choroidal Melanoma Treated With Plaque Radiotherapy

	Case			
	1	2	3	4
Age at presentation, y	48	65	74	77
Race	White	White	White	White
Sex	M	M	M	M
Tumor epicenter	Nasal	Temporal	Inferior	Temporal
Distance to foveola, mm	3.5	0	0	0
Location of anterior margin	Postequatorial	Postequatorial	Equator to ora	Postequatorial
Overhang of optic nerve, %	0	0	0	50
Largest base, mm	9	12	15	10
Thickness, mm	4.2	2.9	4.0	3.4
Shape	Plateau	Dome	Dome	Plateau
Growth pattern	Circumscribed	Diffuse	Diffuse	Circumscribed
Color	Pigmented	Pigmented	Pigmented	Pigmented
Subretinal fluid	Positive	Positive	Positive	Positive
Orange pigment	Positive	Positive	Positive	Positive
Optic disc swelling	Positive	Negative	Negative	Positive
Retinal vascular congestion	Negative	Negative	Negative	Negative
Plaque size, mm	15	15	20	15
Notched, full distribution	Yes	Yes	Yes	Yes
No. of TTT sessions	3	3	2	1
Time to recurrence, mo	10	34	14	12
Treatment of recurrence	TTT (2 sessions)	Enucleation	Enucleation	Enucleation

Abbreviation: TTT, transpupillary thermotherapy.

non-TTT eyes, which could reflect the longer follow-up of 77 months in the non-TTT group and 36 months in the TTT group. Major branch retinal vein occlusion and artery occlusion were encountered more commonly in the TTT group, though no cases of central retinal vascular occlusion were found after treatment.

Long-term visual acuity of 20/200 or worse was measured in 62% overall. In an analysis of 1106 patients with uveal melanoma treated with plaque radiotherapy, poor visual acuity of 20/200 or worse was found in 34% at 5 years and 68% at 10 years.²² In that study, proximity to the foveola and notched-shape plaque were significant risk factors for poor vision and loss of vision. In the current study, visual loss of more than 5 Snellen lines was found in 57% overall, 75% in the non-TTT group, and 46% in the TTT group. Though this may be misinterpreted as a protective effect of TTT, it is much more likely a function of the longer follow-up of the non-TTT group. Possible mechanisms of visual loss following plaque radiotherapy plus TTT include radiation maculopathy, papillopathy, and cataract as well as thermal damage to the optic nerve or foveola, branch vascular obstruction, retinal traction toward the treatment scar, and formation of epiretinal membrane. Although maculopathy, papillopathy, and branch retinal vein occlusion were more frequent in the TTT group, the proportion of patients with long-term visual loss was still greater in the non-TTT group. This was because of higher rates of neovascular complications, cataract, and vitreous hemorrhage in this group. It is likely that some cases in the non-TTT group were planned to receive TTT, but the onset of media opacity did not allow this. Hence, bias may be introduced in the outcomes for the 2 groups, though the disparity in

visual loss from neovascular complications, cataract, and vitreous hemorrhage may also be a function of longer follow-up of the non-TTT group. Of 28 cases that received either TTT or no TTT and no other adjuvant treatment, there were 4 recurrences (14%). In the whole group (n=35) completing at least 3 months' follow-up and including those who had ALPC instead of TTT, this rate falls to 11%. Of 35 eyes, 19 have been followed up for more than 36 months without any recurrences beyond this time, suggesting that tumor recurrence was a relatively early event. In 270 previously reported consecutive cases of all choroidal melanoma managed with plaque radiotherapy combined with TTT, the recurrence rate was 3% at 5 years' follow-up.¹⁴ In circumpapillary tumors, we have found the recurrence rate to be higher. This is a reflection of the posterior juxtapapillary location and the advanced circumferential configuration of the melanoma. Similar recurrence rates of approximately 15% were observed in past studies of plaque radiotherapy in juxtapapillary melanoma.^{11,23} There have been no reports on juxtapapillary melanoma recurrence following charged particle or proton beam radiotherapy, so a direct comparison is not possible.

All 4 cases that had recurrence were in the plaque plus TTT group. This is most likely because these tumors appeared more aggressive at presentation (41% in this group were diffuse melanoma compared with 9% in the non-TTT group) and adjunctive therapy was preplanned on this basis.¹⁴ The limitations of a retrospective series of a rare tumor type may also lead to bias.

Enucleation was necessary in 25% of the 28 cases. The reasons for enucleation included tumor recurrence (3 eyes), neovascular glaucoma (3 eyes), and pain (1 eye).

To the best of our knowledge, no previous reports exist on recurrence or enucleation rates following radiotherapy of circumpapillary melanoma; thus, comparison with results following radiotherapy of all uveal melanoma or specifically juxtapapillary melanoma might serve as a guide but not as a reliable assessment. Globe retention rates have been found to vary between 78% to 94% in studies that include a comparison of helium-ion beam radiotherapy and ^{125}I plaque radiotherapy for all uveal melanomas,²⁴ and proton beam irradiation of macular and juxtapapillary melanomas.⁷ In a past study of juxtapapillary melanoma treated with ^{125}I brachytherapy,¹¹ the enucleation rate was 22%, which is consistent with the findings from the current study. During the study period, there was only 1 case of metastasis (initially treated with plaque only) and no deaths attributable to the primary disease.

In summary, we conclude from the current study that selected eyes with circumpapillary choroidal melanomas that would otherwise be managed with enucleation might be saved with plaque radiotherapy. Based on custom radioactive plaque design and precise plaque placement, tumor control was achieved in 89%. Despite tumor control, visual loss due to tumor-related features and radiation complications led to poor vision in 62% of eyes. Long-term monitoring of these patients for local recurrence and systemic metastasis is advised.

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